

OUEDRAOGO Sayouba

Engineering Student · AI, Automation & Software Development

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PROFILE

Engineering student in Networks & Information Systems with hands-on experience building AI-powered systems, full-stack platforms and backend services. Focused on Deep Learning, LLM agents and production-ready AI pipelines — from model inference (PyTorch, GradCAM) to RAG architectures (LangChain, GPT-4o) and real-time APIs (FastAPI, WebSocket). Curious about ML applied to simulation, HPC workloads and intelligent automation at scale.

TECHNICAL SKILLS

Languages	Python, JavaScript/TypeScript, Java, PHP, Kotlin
AI / LLM	OpenAI API, LangChain, RAG, Whisper, prompt engineering
Backend	FastAPI, Spring Boot, Laravel, REST APIs, WebSocket
Frontend / Mobile	React, Next.js, React Native, Flutter
Data / Infra	PostgreSQL, MySQL, Firebase, Redis, Docker, Git/GitHub

PROJECTS

Smart Attend

University attendance system using facial recognition with real-time notifications and history dashboards for students and teachers.
Spring Boot · FastAPI · React · React Native · PostgreSQL · Redis · InsightFace · Docker · WebSocket

Role: Full-stack development — backend API design, facial recognition pipeline integration and real-time WebSocket events.

Smart Recruit (Hackathon Regional Winner — 1st Place)

AI recruitment microservices platform: automatic CV analysis, compatibility scoring, real-time audio interviews with transcription (Whisper) and LLM-based evaluation (GPT-3.5).

FastAPI · React · TypeScript · MySQL · OpenAI · WebSocket · Docker

Role: Backend lead — microservice architecture, OpenAI integration, real-time audio pipeline and candidate scoring engine.

Smart Learning

Multi-role e-learning platform (student / teacher / admin) with video courses, timed quizzes, PDF certificates, online payment and AI-powered recommendations.

Laravel 11 · PHP · MySQL · Redis · Livewire · Tailwind CSS · OpenAI · PayPal · Docker

Role: Student and teacher dashboards, PayPal payment integration and course access control.

Glaucoma Detection

Medical AI application detecting glaucoma from retinal images with GradCAM visualisation, clinical PDF reports, patient management and a diagnostic chatbot.

FastAPI · PyTorch · React · Three.js · OpenCV · SQLite · Docker

Role: AI backend (PyTorch inference, GradCAM, REST API) and frontend data visualisation.

AI RAG Agent

RAG application to query documents (PDF, Word, CSV) in natural language with sourced answers, conversation history and streaming responses.

FastAPI · React · LangChain · ChromaDB · OpenAI GPT-4o

Role: Personal project — full system design and implementation, from document ingestion to streaming chat interface.

Chat App — Ping Me

Android instant messaging app with photo sharing, friend management, groups, online status and push notifications.

Java · Firebase (Auth, Firestore, Storage, FCM) · Material Design 3

Role: Backend (Java) — Firebase integration, real-time messaging and push notification system.

EDUCATION

2024 – Present	Engineering Degree – Networks & Information Systems · FST Marrakech
2022 – 2024	Diploma of Studies in Sciences and Techniques · FST Settat
2019 – 2022	Technical Baccalaureate (Mathematics, Physical Sciences) · LSN, Bobo Dioulasso

HACKATHONS & COMPETITIONS

Hackathon Ramadan IA	1 st Place — AI recruitment platform (Smart Recruit)
HackShield — SecDojo & AXA GBS	Top 10 / 200 teams — cybersecurity infrastructure challenge

CERTIFICATIONS & LANGUAGES

Certifications: Cisco – Networking Basics | Cisco – Network Administration & Troubleshooting

Languages: French (C1 – Fluent) | English (B1 – Intermediate)